

# An $L^\infty$ -Coefficient Fredholm Alternative for the Periodic-Dirichlet Problem

Candidate solution packet for arXiv:math/0608461

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## Status

**Candidate full solution, likely valid subject to human review.** This packet claims that the open question in Remark 14 of Kmit–Recke [1] has an affirmative answer in the original two-equation, time-independent setting: the hypothesis  $b, c \in BV(0, 1)$  can be replaced by  $b, c \in L^\infty(0, 1)$ . The proof does not address the separate time-dependent coefficient question in Remark 15.

## Source Metadata

- Irina Kmit and Lutz Recke, “Fredholm Alternative for Periodic-Dirichlet Problems for Linear Hyperbolic Systems,” arXiv:math/0608461, J. Math. Anal. Appl. 335 (2007), 355–370 [1].
- Source question location: printed page 18, Section 6, Remark 14.
- Local source PDF: `source_paper.pdf`.

## Question Restated

In the notation of [1], Theorem 1 assumes  $\gamma \geq 1$ ,  $a, d \in L^\infty(0, 1)$ ,  $b, c \in BV(0, 1)$ , and the nonresonance condition

$$|r_0 r_1| \neq \exp \left( \int_0^1 (a(x) + d(x)) dx \right).$$

It then proves that  $A + B : V^\gamma(r_0, r_1) \rightarrow W^\gamma$  is Fredholm of index zero and gives the Fredholm alternative range description via the homogeneous adjoint problem. Remark 14 asks whether the  $BV$  assumption on  $b, c$  can be weakened to  $b, c \in L^\infty(0, 1)$ .

## Proof Intuition

Kmit–Recke prove Fredholmness by applying their abstract lemma to  $C = BA^{-1}$ : it suffices to show that  $C^2$  is compact on  $W^\gamma$ . The compactness criterion requires uniform smallness of translated Fourier modes. All parts of their proof already use only  $L^\infty$  bounds except one high-frequency term, where  $c \in BV$  is used to integrate by parts and obtain a  $1/(1 + |k|)$  decay factor.

That much decay is stronger than the compactness argument needs. It is enough that the troublesome coefficient tends to zero as  $|k| \rightarrow \infty$ . After one application of  $A^{-1}$ , the diagonal solution

## 6 Closing remarks and open questions

In this final section we formulate some closing remarks, generalizations and open questions related to Theorem 1.

**Remark 14 about  $L^\infty$  perturbations of  $b$  and  $c$**  *It seems to be an open question if the assumption  $b, c \in BV(0, 1)$  of Theorem 1 can be weakened to  $b, c \in L^\infty(0, 1)$ . But at least for “almost all”  $b, c \in L^\infty(0, 1)$  Theorem 1 remains to be true. More exactly, the following generalization of assertion (ii) of Theorem 1 holds:*

*Let  $\gamma \geq 1$ ,  $a, d \in L^\infty(0, 1)$ , and suppose (1.7). Then there exists an open and dense set  $M \subseteq (L^\infty(0, 1))^2$  such that  $(BV(0, 1))^2 \subset M$  and that for all  $(b, c) \in M$  the operator  $A + B$  is Fredholm of index zero from  $V^\gamma(r_0, r_1)$  into  $W^\gamma$ .*

*This generalization is true because the set of index zero Fredholm operators is open in  $\mathcal{L}(V^\gamma(r_0, r_1); W^\gamma)$ , because the operator  $B$  depends continuously (in the operator norm in  $\mathcal{L}(W^\gamma)$ ) on the coefficient functions  $b$  and  $c$  (in the  $L^\infty$  norm), and because  $BV(0, 1)$  is dense in  $L^\infty(0, 1)$ .*

Figure 1: The open  $L^\infty$ -coefficient question in Remark 14 of [1].

factors  $e^{ik\omega x + \alpha(x)}u_k(x)$  and  $e^{-ik\omega x - \delta(x)}v_k(x)$  are uniformly bounded in  $H^1(0, 1)$ , after normalization by  $\|f_k\|_2 + \|g_k\|_2$ . Since bounded  $H^1$ -sets are compact in  $C[0, 1]$ , oscillatory integrals against an arbitrary  $L^\infty$  coefficient satisfy a uniform Riemann–Lebesgue lemma on exactly the needed family. This replaces the  $BV$ -based  $1/(1 + |k|)$  estimate by a decay factor  $\mu_k \rightarrow 0$ , and the same finite-tail argument proves compactness of  $C^2$ .

## Main Result

**Theorem 1.** *Let  $\gamma \geq 1$ ,  $a, b, c, d \in L^\infty(0, 1)$ , and assume*

$$|r_0 r_1| \neq \exp \left( \int_0^1 (a(x) + d(x)) dx \right).$$

*Then, in the spaces and notation of [1],*

$$A + B : V^\gamma(r_0, r_1) \longrightarrow W^\gamma$$

*is Fredholm of index zero. Moreover,*

$$\text{im}(A + B) = \left\{ (f, g) \in W^\gamma : \int_0^{2\pi/\omega} \int_0^1 (fu + gv) dx dt = 0 \text{ for all } (u, v) \in \ker(\tilde{A} + \tilde{B}) \right\}.$$

*Thus all assertions of Theorem 1 of [1] remain true when the hypothesis  $b, c \in BV(0, 1)$  is replaced by  $b, c \in L^\infty(0, 1)$ .*

## Proof

We follow the notation of [1]. Put

$$\alpha(x) = \int_0^x a(y) dy, \quad \delta(x) = \int_0^x d(y) dy,$$

and

$$\Delta_k = e^{ik\omega + \delta(1)} - r_0 r_1 e^{-ik\omega - \alpha(1)}.$$

The nonresonance condition gives  $\inf_k |\Delta_k| > 0$ , exactly as in [1, (3.8)].

**Lemma 1** (uniform oscillation on  $H^1$ -bounded sets). *Let  $h \in L^\infty(0, 1)$  and  $R > 0$ . Then*

$$\eta_m(h, R) := \sup_{\|q\|_{H^1(0,1)} \leq R} \sup_{0 \leq x \leq 1} \left| \int_0^x e^{im\omega y} h(y) q(y) dy \right| \rightarrow 0$$

as  $|m| \rightarrow \infty$ .

*Proof.* The closed  $H^1$ -ball of radius  $R$  is compact in  $C[0, 1]$ . Given  $\varepsilon > 0$ , choose a finite  $C[0, 1]$ -net  $q_1, \dots, q_N$  for this ball with mesh  $\varepsilon/(3\|h\|_\infty + 1)$ . For each fixed  $q_j$ , the function  $hq_j$  lies in  $L^1(0, 1)$ , and the indefinite Riemann–Lebesgue lemma gives

$$\sup_{0 \leq x \leq 1} \left| \int_0^x e^{im\omega y} h(y) q_j(y) dy \right| \rightarrow 0.$$

For completeness, this indefinite version follows by approximating  $hq_j$  in  $L^1$  by a step function: step functions have uniformly  $O(|m|^{-1})$  oscillatory primitives, and the  $L^1$ -error bounds the primitive error uniformly in  $x$ . Taking the maximum over the finite net and then using the mesh bound proves the claim.  $\square$

**Lemma 2** (the  $BV$  estimate has an  $L^\infty$  replacement). *Let  $(u_k, v_k) = A^{-1}(f_k, g_k)$  be the modewise solution given by [1, (3.6)], and set*

$$E_k = \|f_k\|_{L^2(0,1)} + \|g_k\|_{L^2(0,1)}.$$

*If  $b, c \in L^\infty(0, 1)$ , then there is a sequence  $\mu_k \rightarrow 0$  as  $|k| \rightarrow \infty$ , depending only on the coefficients and boundary constants, such that, for every  $x \in [0, 1]$ ,*

$$\left| \int_0^x e^{-ik\omega y - \delta(y)} c(y) u_k(y) dy \right| \leq \mu_k E_k,$$

$$\left| \int_0^x e^{ik\omega y + \alpha(y)} b(y) v_k(y) dy \right| \leq \mu_k E_k,$$

and

$$|w_k(bv_k, cu_k)| \leq \mu_k E_k,$$

where  $w_k$  is the functional in [1, (3.7)].

*Proof.* Define

$$q_k(x) := e^{ik\omega x + \alpha(x)} u_k(x), \quad s_k(x) := e^{-ik\omega x - \delta(x)} v_k(x).$$

From the explicit diagonal formula [1, (3.6)],

$$q'_k(x) = e^{ik\omega x + \alpha(x)} f_k(x), \quad s'_k(x) = e^{-ik\omega x - \delta(x)} g_k(x),$$

and the initial constants  $q_k(0)$ ,  $s_k(0)$  are bounded by  $CE_k$ , using the uniform lower bound on  $|\Delta_k|$ . Hence

$$\|q_k\|_{H^1(0,1)} + \|s_k\|_{H^1(0,1)} \leq CE_k.$$

If  $E_k = 0$ , the desired estimates are trivial. Otherwise apply Lemma 1 to the  $H^1$ -bounded functions  $q_k/E_k$  and  $s_k/E_k$ . Since

$$\int_0^x e^{-ik\omega y - \delta(y)} c(y) u_k(y) dy = \int_0^x e^{-2ik\omega y} c(y) e^{-\alpha(y) - \delta(y)} q_k(y) dy$$

and

$$\int_0^x e^{ik\omega y + \alpha(y)} b(y) v_k(y) dy = \int_0^x e^{2ik\omega y} b(y) e^{\alpha(y) + \delta(y)} s_k(y) dy,$$

both displayed integrals are bounded by  $o_{|k| \rightarrow \infty}(1)E_k$ , uniformly in  $x$ . The estimate for  $w_k(bv_k, cu_k)$  follows from the same two global integrals at  $x = 1$  and the bounded scalar factors in the definition of  $w_k$ .  $\square$

Now let  $N \subset W^\gamma$  be bounded and put

$$M = (BA^{-1})^2 N.$$

As in Section 4 of [1], it is enough to verify the componentwise compactness criterion of [1, Lemma 2]. Uniform boundedness of  $M$  follows from boundedness of  $B$  on  $W^\gamma$  and of  $A^{-1} : W^\gamma \rightarrow V^\gamma(r_0, r_1)$ , both valid under  $L^\infty$  coefficient assumptions.

For the shift-continuity condition, repeat the proof of [1, Section 4]. The first shift term is estimated by

$$C\xi^2$$

using only  $b, c \in L^\infty$  and the uniform mode estimate [1, (3.9)]. The third shift term tends to zero by  $L^2$  continuity in the mean of  $x \mapsto b(x)e^{\delta(x)}$ , again requiring only  $b \in L^\infty$ . The identical statements with  $b$  and  $c$  interchanged handle the  $v$ -component.

It remains only to replace the estimate of the middle term [1, (4.10)]. By Lemma 2 and  $\inf_k |\Delta_k| > 0$ , the expression inside the last absolute value in [1, (4.10)] is bounded by  $\mu_k E_k$ , with  $\mu_k \rightarrow 0$ . Therefore that middle contribution is at most

$$C \sum_{k \in \mathbb{Z}} (1 + k^2)^\gamma |e^{ik\omega(\xi + \tau)} - 1|^2 \mu_k^2 \left( \|f_k\|_{L^2(0,1)}^2 + \|g_k\|_{L^2(0,1)}^2 \right),$$

where  $C$  is independent of  $(f, g) \in N$ ,  $\xi$ , and  $\tau$ . Since  $N$  is bounded in  $W^\gamma$ , given  $\varepsilon > 0$  choose  $K$  so large that  $\sup_{|k| > K} \mu_k^2$  makes the tail  $< \varepsilon/2$ , using  $|e^{i\theta} - 1| \leq 2$ . The finite sum over  $|k| \leq K$  tends to zero as  $\xi, \tau \rightarrow 0$ , uniformly for  $(f, g) \in N$ , because there are only finitely many factors  $e^{ik\omega(\xi + \tau)} - 1$  and the weighted mode norms are uniformly bounded on  $N$ . Thus the middle term also tends to zero uniformly.

The compactness criterion gives that  $(BA^{-1})^2$  is compact on  $W^\gamma$ . Kmit-Recke's abstract Lemma 11, which states that  $I + C$  is Fredholm when  $C^2$  is compact, applied to  $C = BA^{-1}$ , shows that  $I + BA^{-1}$  is Fredholm. Therefore  $A + B$  is Fredholm.

It remains to record that the index-zero and range-identification arguments in Section 5 of [1] do not use derivatives of  $b$  or  $c$ . They use the adjoint identity for  $L^\infty$  coefficients, the Fredholmness just proved, and the standard finite-dimensional equality between the number of solutions of a modewise  $2 \times 2$  linear boundary-value ODE and the number of solutions of its adjoint boundary-value problem. These arguments therefore carry over verbatim to  $b, c \in L^\infty(0, 1)$ . Hence  $A + B$  has index zero and the asserted Fredholm alternative range description follows.  $\square$

## Verification Notes

- Local duplicate search: the lightweight indexes `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv` were searched for arXiv id 0608461, the title, “periodic-Dirichlet”, and  $BV/L^\infty$  Fredholm keywords. No prior exact packet was found.
- Source proof audit: The only located use of  $b, c \in BV(0, 1)$  in the Fredholmness proof is the integration-by-parts estimate immediately after [1, (4.10)]. Lemma 2 replaces exactly this estimate.
- The proof is analytic; no numerical verification is relevant. The compactness step is the main item for human review.

## Novelty Check

Bounded literature checks were performed on 2026-07-18. The arXiv record for the source paper identifies the published version and DOI [1]. Kmit’s 2007 follow-up paper on the whole Sobolev scale treats  $L^\infty$  coefficients for a priori estimates and operator representation, but its reference list still points to Kmit–Recke’s  $BV$ -coefficient Fredholm alternative rather than resolving the arbitrary  $L^\infty$  Fredholm question [2]. Kmit–Recke’s later 2012 general  $n \times n$  theory gives broader Fredholm criteria under structural assumptions, but the search found no statement that directly answers the special two-speed arbitrary- $L^\infty$  question of Remark 14 [3]. Thus the packet is promoted as a new candidate proof, not as a literature-status identification. Novelty confidence should still be treated as bounded rather than exhaustive.

## Human Review Recommendation

Review Lemma 1 and the normalization step in Lemma 2, then compare the final finite-tail estimate with the source proof of [1, (4.10)]. If these checks pass, the packet appears to give the desired full  $L^\infty$  strengthening of Remark 14.

## References

- [1] I. Kmit and L. Recke, *Fredholm Alternative for Periodic-Dirichlet Problems for Linear Hyperbolic Systems*, J. Math. Anal. Appl. 335 (2007), 355–370. arXiv:math/0608461. DOI: 10.1016/j.jmaa.2007.01.055.
- [2] I. Kmit, *Linear hyperbolic problems in the whole scale of Sobolev-type spaces of periodic functions*, Comment. Math. Univ. Carol. 48 (2007), no. 4, 631–645. EUDML record.

- [3] I. Kmit and L. Recke, *Fredholmness and smooth dependence for linear time-periodic hyperbolic systems*, J. Differential Equations 252 (2012), no. 2, 1962–1986. DOI: 10.1016/j.jde.2011.09.013.